

3.2 Hydrocarbons

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.2 Hydrocarbons
Difficulty	Easy

Time allowed: 40

Score: /27

Percentage: /100

Question 1a

State **two** reasons why alkanes are unreactive.

[2 marks]

Question 1b

Methane reacts with chlorine to produce chloromethane, CH_3Cl . The reaction is initiated by the formation of chlorine radicals.

i)

State what is meant by a radical.

[1]

ii)

State the condition required to form chlorine free radicals from Cl_2 .

[1]

[2 marks]

Question 1c

State the type of bond fission involved in the reaction in part **(b)**.

[1 mark]

Question 1d

Chloromethane will be formed via several steps.

i)

Write the equations for the **two** propagation steps.

[2]

ii)

Write an equation to show how ethane can be formed in this reaction.

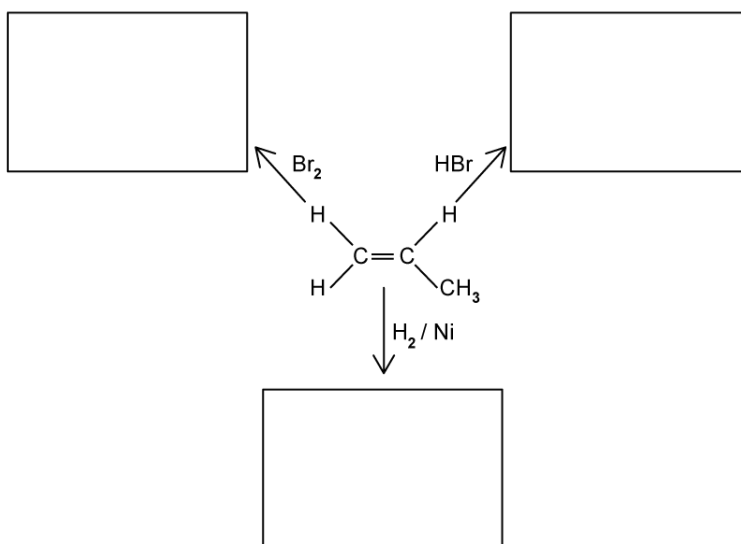
[1]

[3 marks]

Question 2a

This question is about the reactions of propene.

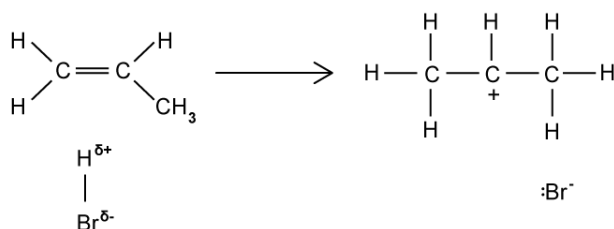
Complete the flowchart below to show the **major** organic product formed in each of the addition reactions.



[3 marks]

Question 2b

Use 'curly' arrows to complete the mechanism for the formation of the major product when propene CH_3CHCH_2 , reacts with hydrogen bromide, HBr .



[2 marks]

Question 2c

Draw the displayed formula of the minor product formed from the reaction of propene and hydrogen bromide.

[1 mark]

Question 2d

Explain why there is a major and minor product formed from the reaction between propene and hydrogen bromide.

[3 marks]

Question 3a

Alkanes can undergo both complete and incomplete combustion.

State the difference between complete and incomplete combustion of a hydrocarbon fuel, such as octane. Include the balanced symbol equation for each in your answer.

[4 marks]

Question 3b

Octane is found in petrol.

Carbon monoxide is a colourless, odourless and poisonous gas which is emitted in the exhaust fumes of motor vehicles that use a petrol internal combustion engine.

i)
State why carbon monoxide can cause dizziness, fainting, loss of consciousness and death in humans.

[1]

ii)
State another pollutant that is also released in cars' exhaust fumes. Explain the effect of this pollutant in the atmosphere.

[2]

[1 mark]

Question 3c

Octane can be cracked to form pentane and one other product.

i)

Give the balanced symbol equation for this reaction.

[1]

ii)

Describe how catalytic cracking is carried out.

[2]

[2 marks]

Question 3d

Describe and give the result of a chemical test that would distinguish between the two products.

[3 marks]

Model Answers: Easy

1a

a) Alkanes are unreactive due to:

- The high bond enthalpies / strength of the C-H bonds (and C-C bonds); [1 mark]
- The very low polarity of the C-H / sigma (σ) bonds present; [1 mark]

[Total: 2 marks]

- The C-C bond has a bond enthalpy of 346 kJ mol^{-1} and the C-H bond has a bond enthalpy of 411 kJ mol^{-1}
- The electronegativities of carbon and hydrogen are very similar, therefore the bonds will have very low polarity
- This means we need specific conditions in order to get alkanes to react

1b

b)

i) A radical is:

- A species with an unpaired electron; [1 mark]

ii) The condition required to form chlorine free radicals from Cl_2 is:

- Ultraviolet light; [1 mark]

[Total: 2 marks]

- A radical is represented by an $\text{X}\cdot$ on the atom with an unpaired electron

1c

c) The type of bond fission is:

- Homolytic fission; [1 mark]

[Total: 1 mark]

- Homolytic fission is breaking a covalent bond in such a way that each atom takes an electron from the bond to form two radicals
- The movement of electrons are shown by single headed arrows
- This is because one electron is being moved to each atom



1d

d)

i) The two propagation steps are:

- $\text{Cl}\cdot + \text{CH}_4 \rightarrow \text{CH}_3\cdot + \text{HCl}$; [1 mark]
- $\text{Cl}_2 + \text{CH}_3\cdot \rightarrow \text{CH}_3\text{Cl} + \text{Cl}\cdot$; [1 mark]

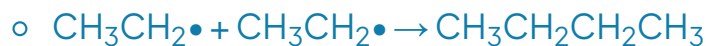
ii) Ethane can be formed by:

- $\text{CH}_3\cdot + \text{CH}_3\cdot \rightarrow \text{CH}_3\text{CH}_3 / \text{C}_2\text{H}_6$
OR
 $\text{CH}_3\cdot + \text{CH}_3\cdot \rightarrow \text{C}_2\text{H}_6$; [1 mark]

[Total: 3 marks]

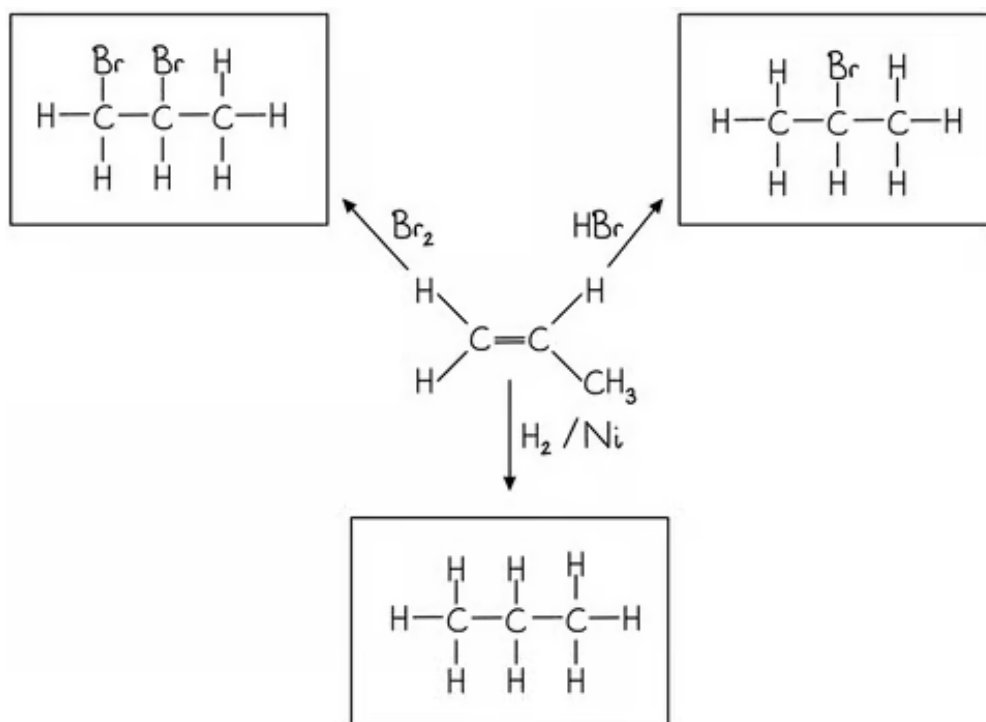
- During both propagation steps a free radical must be used and generated
- Ethane can be formed via one of the possible termination steps involving the net destruction of two free radicals
- This is actually a limitation of the reaction as we are forming an undesired product
- Possible termination steps in this reaction are:
 - $\text{CH}_3\cdot + \text{CH}_3\cdot \rightarrow \text{CH}_3\text{CH}_3$
 - $\text{CH}_3\cdot + \text{Cl}\cdot \rightarrow \text{CH}_3\text{Cl}$
 - $\text{Cl}\cdot + \text{Cl}\cdot \rightarrow \text{Cl}_2$
- **Careful:** If you are asked to show the formation of an alkane with a longer chain

such as butane, make sure you write



2a

a) The organic product formed in each reaction are as follows:



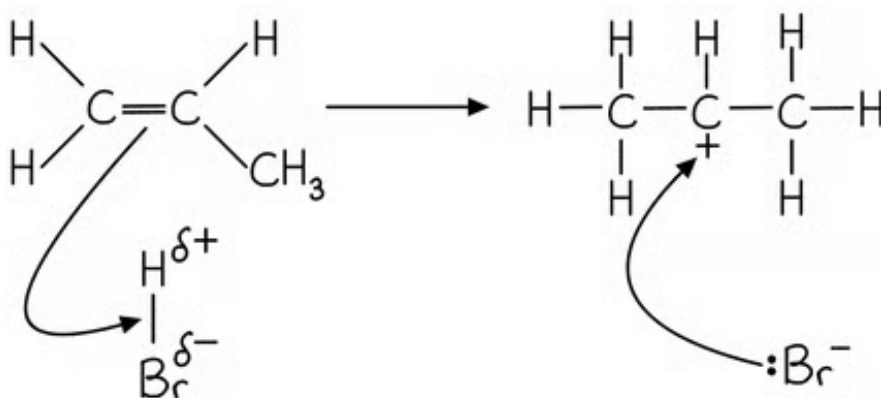
- Each correct structure; [1 mark]

[Total: 3 marks]

- The question asks for the major product
- This does not influence the products for the reaction with H_2 / Ni and Br_2 but does for HBr

2b

b) The curly arrows should be added as follows:



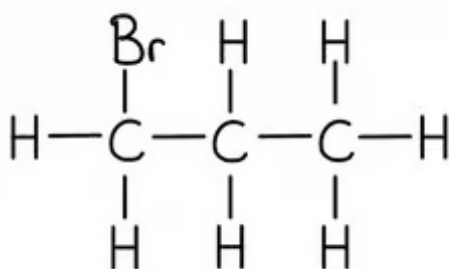
- Curly arrow from C=C to H-Br bond; [1 mark]
- Curly arrow from lone pair on Br⁻ ion to carbocation; [1 mark]

[Total: 2 marks]

- **Remember:** Curly arrows must start from an area of negative charge and go to an area of positive charge
- You could also be asked to add in the δ^+ and δ^- charges to the H-Br molecule so double-check the question
 - If the question is not clear about this, then it is always better to add them just in case

2c

c) The displayed formula of the minor product is:



- Correct structure of 1-bromopropane; [1 mark]

[Total: 1 mark]

- **Remember:** The addition of HBr and similar chemicals to unsymmetrical alkenes will result in the formation of a major and minor product

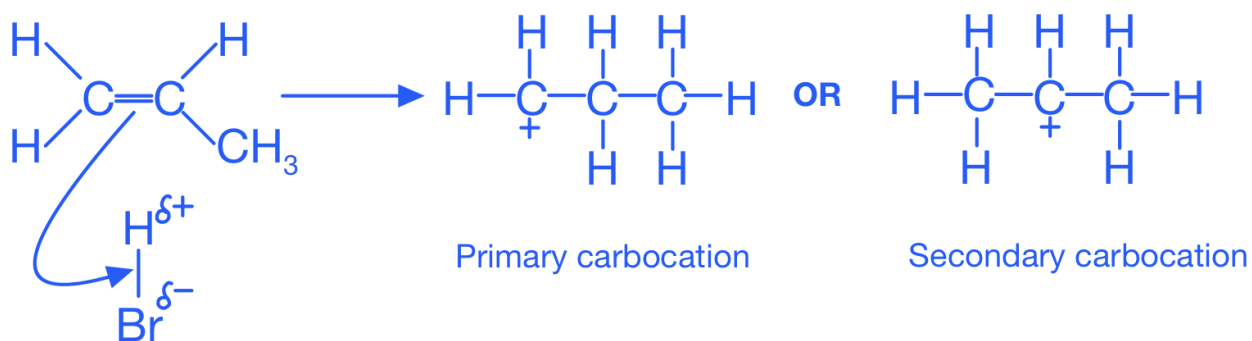
2d

d) A major and a minor product are produced because:

- Intermediate can be a primary or secondary carbocation; [1 mark]
- Secondary carbocation is more stable than primary carbocation; [1 mark]
- Major product formed from secondary carbocation; [1 mark]

[Total: 3 marks]

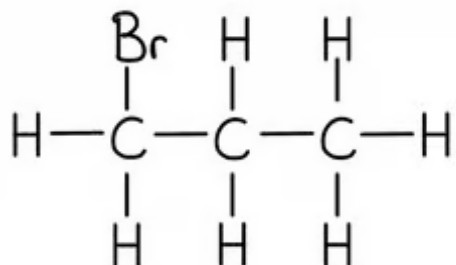
- You would be awarded the marks if you had given the reverse argument for the primary carbocation being less stable and forming the minor product
- The two possible intermediates are:



case

2c

c) The displayed formula of the minor product is:



- Correct structure of 1-bromopropane; [1 mark]

[Total: 1 mark]

- **Remember:** The addition of HBr and similar chemicals to unsymmetrical alkenes will result in the formation of a major and minor product

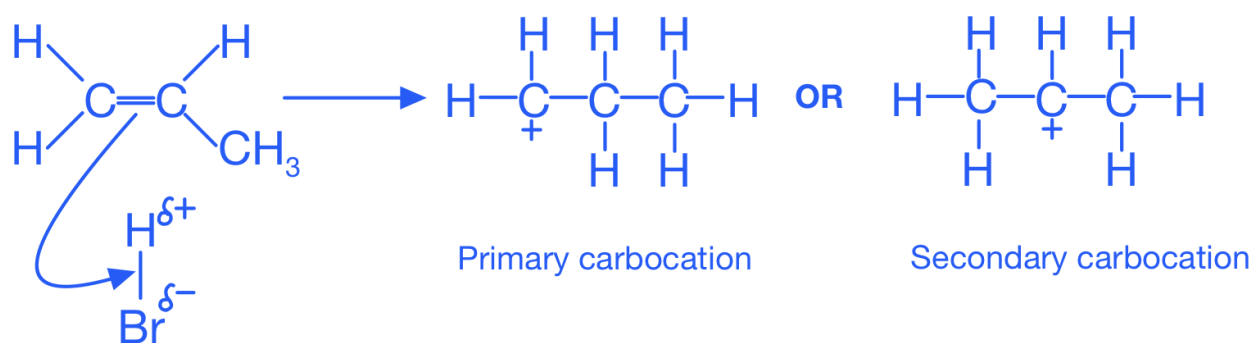
2d

d) A major and a minor product are produced because:

- Intermediate can be a primary or secondary carbocation; [1 mark]
- Secondary carbocation is more stable than primary carbocation; [1 mark]
- Major product formed from secondary carbocation; [1 mark]

[Total: 3 marks]

- You would be awarded the marks if you had given the reverse argument for the primary carbocation being less stable and forming the minor product
- The two possible intermediates are:



- As shown in the mark scheme for part **(b)**, the second step of the mechanism is for the bromide ion, :Br^- to attack the carbocation
- The secondary carbocation is more stable than the primary carbonation
- This means that is more likely to form resulting in the major product, 2-bromopropane

3a

a) The difference between complete and incomplete combustion of a hydrocarbon fuel such as octane, including the balanced symbol equation, is:

- Complete combustion has an excess / plentiful supply of oxygen; [1 mark]
- $\text{C}_8\text{H}_{18} + 12.5\text{O}_2 \rightarrow 8\text{CO}_2 + 9\text{H}_2\text{O}$; [1 mark]
- Incomplete combustion has a limited / reduced supply of oxygen; [1 mark]
- $\text{C}_8\text{H}_{18} + 8.5\text{O}_2 \rightarrow 8\text{CO} + 9\text{H}_2\text{O}$

OR**[Total: 4 marks]**

- When a fuel is burnt it can undergo complete or incomplete combustion
 - The determining factor is the amount of oxygen
- Complete combustion of a hydrocarbon produces carbon dioxide and water
- To balance the complete combustion of octane equation:
 - Balance carbon atoms
 - $\text{C}_8\text{H}_{18} + \text{O}_2 \rightarrow 8\text{CO}_2 + \text{H}_2\text{O}$
 - Balance hydrogen atoms
 - $\text{C}_8\text{H}_{18} + \text{O}_2 \rightarrow 8\text{CO}_2 + 9\text{H}_2\text{O}$
 - Balance oxygen atoms

- $\text{C}_8\text{H}_{18} + 12.5\text{O}_2 \rightarrow 8\text{CO}_2 + 9\text{H}_2\text{O}$
- You could also double the balancing numbers to give whole numbers:
 - $2\text{C}_8\text{H}_{18} + 25\text{O}_2 \rightarrow 16\text{CO}_2 + 18\text{H}_2\text{O}$
- Incomplete combustion of a hydrocarbon can produce either carbon monoxide and water or carbon and water
- You are expected to know the equation for the production of carbon monoxide and water
- To balance the incomplete combustion of octane equation:
 - Balance carbon atoms
 - $\text{C}_8\text{H}_{18} + \text{O}_2 \rightarrow 8\text{CO} + \text{H}_2\text{O}$
 - Balance hydrogen atoms
 - $\text{C}_8\text{H}_{18} + \text{O}_2 \rightarrow 8\text{CO} + 9\text{H}_2\text{O}$
 - Balance oxygen atoms
 - $\text{C}_8\text{H}_{18} + 8.5\text{O}_2 \rightarrow 8\text{CO} + 9\text{H}_2\text{O}$
 - You could also double the balancing numbers to give whole numbers:
 - $2\text{C}_8\text{H}_{18} + 17\text{O}_2 \rightarrow 16\text{CO} + 18\text{H}_2\text{O}$

3b

b)

i) Carbon monoxide cause dizziness, fainting, loss of consciousness and death in humans by:

Any **one** of the following:

- Reduces ability of red blood cells to carry oxygen; [1 mark]
- Binding irreversibly / very tightly to haemoglobin / red blood cells; [1 mark]
- Stopping oxygen attaching to haemoglobin / red blood cells; [1 mark]
- Forms carboxyhaemoglobin instead of oxyhaemoglobin; [1 mark]

ii) Another pollutant that is also released in cars' exhaust fumes and the effect of this pollutant in the atmosphere is:

EITHER

- Oxides of nitrogen / NO_x / nitrogen monoxide / NO / nitrogen dioxide / NO_2 ;
- React with water to form nitric acid / acid rain;

OR

Reacts with VOCs / unburnt hydrocarbons to form PAN / peroxyacetyl nitrate /

photochemical smog;

OR

- Unburnt hydrocarbons / VOCs;
- Reacts with oxides of nitrogen / NO_x / nitrogen monoxide / NO / nitrogen dioxide / NO_2 to form PAN / peroxyacetyl nitrate / photochemical smog;

[Total: 3 marks]

- Carbon monoxide is known as the silent killer because it is colourless, odourless and poisonous
- You need to know the environmental consequences of carbon monoxide, oxides of nitrogen and unburnt hydrocarbons arising from the combustion of alkanes in the internal combustion engine

3c

c)

i) The balanced symbol equation for the reaction is:

- $\text{C}_8\text{H}_{18} \rightarrow \text{C}_5\text{H}_{12} + \text{C}_3\text{H}_6$; [1 mark]

ii) Catalytic cracking is carried out by:

- Heating the hydrocarbon to a high temperature; [1 mark]
- Passing the vapours over aluminium oxide / Al_2O_3 ; [1 mark]

[Total: 2 marks]

- When writing an equation for cracking, the number of carbon and hydrogen atoms need to be balanced on each side of the equation
- Cracking of alkanes produces a smaller alkane and alkene
- You are told that pentane is formed, which you should know the formula of
- It is an alkane that has 5 carbon atoms and obeys the general equation $\text{C}_n\text{H}_{2n+2}$, so its formula is C_5H_{12}
 - $\text{C}_8\text{H}_{18} \rightarrow \text{C}_5\text{H}_{12} +$
- The other product must account for the remaining C and H atoms, therefore it must have 3 C atoms and 6 H atoms, so C_3H_6
 - As expected, this obeys the general formula of an alkene, C_nH_{2n}

- This gives the equation:
 - $\text{C}_8\text{H}_{18} \rightarrow \text{C}_5\text{H}_{12} + \text{C}_3\text{H}_6$
- Cracking is a thermal decomposition reaction
- The long chain alkane is broken down using heat
- Just stating heat on its own is insufficient, you must say at high temperatures
- Other catalysts that would also be accepted are:
 - Alumina **or** zeolites **or** aluminosilicates **or** broken / porous pot

3d

d) A chemical test and result that would distinguish between the two products would be:

- Add bromine water / $\text{Br}_2(\text{aq})$; [1 mark]
- Pentane / C_5H_{12} would remain orange; [1 mark]
- Propene / C_3H_6 would go colourless; [1 mark]

[Total: 3 marks]

- The alkene reacts because it contains a $\text{C}=\text{C}$ double bond, and the bromine can add across it
 - This is called an addition reaction

